The Log4j properties file (log4j2.properties) consists of system variables (`sys:es.logs.base_path`, `sys:es.logs.cluster_name`, etc.) that are resolved at application runtime. Because Elasticsearch exposes these properties, they are available to Log4j, which lets Log4j set up its log file directory location, log file pattern, and other properties. For example, `sys:es.logs.base_path` points to the path where Elasticsearch writes the logs, which resolves to the $ES_HOME/logs directory.

By default, most of Elasticsearch runs at the `INFO` level, but we can customize the setting based on individual packages. For example, we can edit the log4j2.properties file and add a logger for the `index` package, as the following listing shows.

Listing 14.13 Setting the logging level for a specific package

```
logger.index.name =  org.elasticsearch.index
logger.index.level =  DEBUG
```

Doing this allows the `index` package to output logs at the `DEBUG` level. Rather than editing this file on a specific node and restarting that node (we may need to do this for every node if we haven't done so before creating a farmed cluster), we can set the `DEBUG` log level at the cluster level for this package.

Listing 14.14 Setting the transient log level globally

```
PUT _cluster/settings
{
  "transient": {
    "logger.org.elasticsearch.index":"DEBUG"
  }
}
```

The transient setting

Sets this index package's logging level to DEBUG

As the query shows, we set the logger-level property for the `index` package to `DEBUG` in the `transient` block. The `transient` block indicates that the property is not durable (only available while the cluster is up and running). If we restart the cluster or it crashes, the setting is lost because it is not stored permanently on the disk.

We can set this property with a call to the cluster settings API (`_cluster/settings`), as listing 14.14 shows. Once this property is set, any further logging info related to the `index` in the `org.elasticsearch.index` source package is output at the `DEBUG` level.

Elasticsearch also provides a means of storing cluster properties durably. To store the properties permanently, we can use the `persistent` block. The following listing replaces the `transient` block with a `persistent` block.

Listing 14.15 Setting the log level permanently

```
PUT _cluster/settings
{
  "persistent": {
    "logger.org.elasticsearch.index":"DEBUG",
```